

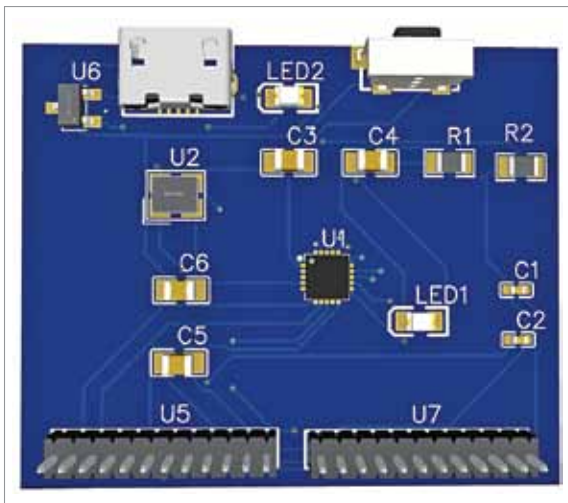


Fig. 6: RISC-V chip based board design

indication of the system's power status, an indicator LED is connected to the power pin of the CH32V003F4U6 microcontroller. This LED lights up when the system is powered, offering a quick and intuitive assessment of the operational status.

For user feedback and debugging purposes, an in-built LED is integrated into the design, connected to pin PC0 of the CH32V003F4U6 microcontroller. This LED can be programmed to convey specific system states or to assist in troubleshooting.

Circuit configuration

The reset button is connected to pin PD7 with a capacitor for debouncing. The indicator LED is connected to the power pin, providing a visual cue when the system is powered. The in-built LED is connected to pin PC0 for programmable feedback. The USB mini female connector is linked to the voltage regulator's Vin pin, ensuring a stable power supply. Fig. 3 shows RISC-V chip circuit configuration

Adding clock

A crystal oscillator is essential for providing precise timing in

microcontroller-based systems. It acts as the heartbeat, ensuring that all operations occur synchronously and consistently. The inclusion of a 24MHz crystal oscillator elevates the accuracy of the system, benefiting applications that demand precise timing and synchronisation.

The 24MHz crystal is connected across the PCO (crystal output) and PCI (crystal input) pins of the CH32V-

003F4U6 microcontroller. This connection establishes a stable clock input, allowing the microcontroller to execute instructions with precision. Two 22pF capacitors are connected from each crystal pin to the ground (GND), providing the necessary load capacitance for the crystal oscillator. Fig. 4 shows how to add the crystal.

Completing the system design

A crucial element for ensuring stable power distribution is introduced in the final stage of the system—the decoupling capacitor. Connecting the decoupling capacitor between the VDD and GND pins of the CH32V003F4U6 microcontroller, enhances the system's reliability and mitigates potential noise interference. With this addition, the RISC-V based system is poised for diverse applications, as sensors, actuators, headers, and connectors can now be seamlessly integrated into the GPIO pins.

1. The role of decoupling capacitor. The decoupling capacitor acts as a buffer, providing a stable and noise-free power supply to the microcontroller. It counteracts voltage fluctuations, ensuring that the microcontroller receives a clean

and consistent power signal, which is critical for reliable operation.

2. Connecting decoupling capacitor. The decoupling capacitor is connected between the VDD and GND pins of the CH32V003F4U6 microcontroller. Typically, a capacitor in the range of a few microfarads is chosen to effectively filter out high-frequency noise and stabilise the power supply.

3. Expanding the system. With the foundational elements in place—the microcontroller, voltage regulator, crystal oscillator, reset button, indicator LED, in-built LED, and now the decoupling capacitor—the system is primed for expansion. Sensors and actuators can be seamlessly integrated into the GPIO pins, and headers/connectors can be added to facilitate communication with external devices.

4. Programming and implementation. The CH32V003F4U6 microcontroller can be programmed according to the specific requirements of the application. The versatile GPIO pins provide the flexibility to interface with a wide array of sensors and control various actuators. With headers and connectors in place, the system can be easily integrated into larger projects or embedded into custom-designed PCBs.

Complete circuit of RISC-V chip

The complete design, including reset, indicators, USB connector, decoupling capacitor, power supply, crystal oscillator, inputs and outputs of the system are shown in Fig. 5, while the PCB for RISC-V chip based board design is shown in Fig. 6.

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/electronics-projects/risc-v-chip> **EFY**

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